

Solution Architecture

Technical Architecture for HealHub

Project	HealHub
Tagline	Care At Your Fingertips
Domain	Healthcare / Doctor Appointment Booking
Stack	MERN Stack
Generated On	13 June 2026

Architecture Diagram - Text Representation

React UI -> api.js fetch layer -> Express.js routes -> Controllers -> Mongoose models -> MongoDB Atlas.
Frontend is deployed on Vercel and backend is deployed on Render.

Architecture Layers

Layer	Component	Responsibility
Presentation	React pages/components	Display UI and collect user input
Routing	React Router	Navigate pages and dashboards
API Client	api.js	Send requests to backend URL
Backend	Express.js	Expose REST endpoints
Business Logic	Controllers/routes	Handle registration, login, doctors, appointments
Database	MongoDB Atlas	Store project data
Deployment	Vercel/Render	Host application online

Environment Configuration

Variable	Purpose
VITE_API_BASE_URL	Frontend backend API URL on Vercel
MONGO_URI	MongoDB Atlas connection string on Render
JWT_SECRET	Token signing secret if JWT authentication is used
PORT	Backend server port